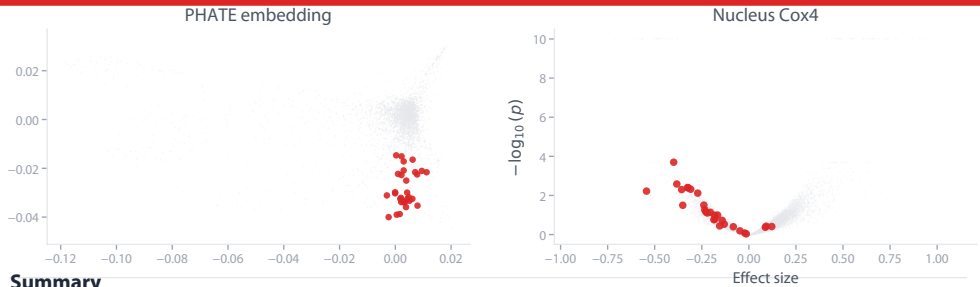


Pre-mRNA splicing (spliceosome assembly and function)

Cluster 5



Summary

Cluster 5 shows HIGH confidence for pre-mRNA splicing as the dominant pathway, with 18/31 genes (58%) being established core spliceosome components representing multiple subcomplexes (U2 snRNP SF3A/SF3B, U4/U6-U5 tri-snRNP, PRP19 complex, LSM proteins). Two completely uncharacterized genes (LOC102724594, ENSG00000289690) represent excellent discovery opportunities as potential novel splicing facto...

Established genes

SMNDC1, PHF5A, PRPF4, CWC22, U2AF2, PRPF18, SF3A2, CDC40, LSM3, SF3B6, PRPF19, SF3A3, PLRG1, SRSF2, EFTUD2, ISY1, LSM5, SF3B1

Novel & uncharacterized genes

PSMD12 *novel*

26S proteasome component involved in protein degradation. While proteasome may regulate splicing factor turnover, this is a general protein homeostasis function. Connection is indirect and likely

USPL1 *novel*

SUMO-specific isopeptidase with established role in snRNA transcription in Cajal bodies and binding to U snRNA genes. Already connected to snRNA biology and splicing machinery, making this

SAP30BP *novel*

Involved in transcriptional repression through histone deacetylase activity. While no direct splicing role documented, connection between transcriptional repression and splicing is well-

KIN *novel*

Involved in DNA replication, DNA damage response, and has documented RNA binding and potential mRNA processing roles. Connection to RNA processing is already suggested in literature, reducing novelty

MEPCE *novel*

Methyltransferase for 7SK snRNA involved in P-TEFb regulation and transcriptional elongation. Also stabilizes U6 snRNA. While U6 is a spliceosomal component, MEPCE's role in spliceosome function is

ECD *novel*

Known regulator of p53 stability and cell cycle. Has documented role in pre-mRNA splicing regulation and mRNA nuclear export. Connection to splicing is already established in literature

BRF2 *novel*

RNA polymerase III transcription factor for genes with upstream promoter elements. No established connection to pre-mRNA splicing (Pol II function). May indicate cross-talk between Pol III and Pol

FAM32A *unc.*

Limited characterization showing potential role in G2 arrest and apoptosis (1 publication). No known connection to splicing. Clustering suggests possible novel role in coordinating splicing with cell

ZNF335 *novel*

Known methyltransferase regulator and component of histone methyltransferase complexes controlling neural progenitor proliferation. No documented splicing function. May represent novel link between

+ 4 more

18 established · 10 novel · 3 uncharacterized